Coronae, pulses and particles: archival tests for artificial emission in stellar activity and high-energy surveys

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Abstract

A technology on or near a star could add X-rays a corona cannot make, radio emission a quiet star should not have, a pulse train, a flare clock, or particles from a direction no astrophysical source occupies. We put each of these to the archive that measures it. Of 137,896 eROSITA coronal counterparts, none exceeds a tenth of its star's luminosity in X-rays, and the 250 above a hundredth are evolved stars, catalogued cataclysmic, young or extragalactic sources and M dwarfs whose flares reach that level; no slow rotator stands far above the rotation-activity relation, and among 63,043 eROSITA point sources none holds its counts in a band where no continuum could put them. No X-ray light curve of a star within 10 to 20 pc holds a coherent period beyond the noise. Radio emission from inactive stars, circular polarisation without a counterpart and NVSS sources polarised beyond the synchrotron ceiling all trace to background galaxies, active stars, pulsars or the diffuse emission of their fields, and no GLEAM-X spectrum steps at a sub-band edge. No neutrino source sits at the 271 systems within 10 pc or moves with Jupiter, Saturn, Mars or Venus, and ultra-high-energy cosmic rays arrive in pairs and triplets as often as chance allows. In photon timing, flares and pulses, every cluster of ultraviolet photons on a nearby star is a flare, no flaring star keeps a clock better than a part in a thousand, planet hosts flare as their matched stars do, and the dispersed pulses of the nearest stars' radio cadences appear as often off the stars.

Keywords: Technosignatures, Stars: activity, X-rays: stars, Radio continuum: stars, Neutrinos, Cosmic rays

1. Introduction

Stars are bright in every band, and a technology among them is most easily hidden in their light. The places where it would show are the places where a star's own physics sets a limit: a corona saturates at about a thousandth of the star's luminosity [1, 2]; a quiet star's radio emission is faint and unpolarised; synchrotron radiation polarises at most 72 per cent; flares come at random times; and neutrinos and the highest-energy cosmic rays come from cosmic accelerators, not from nearby stars. A transmitter, a power plant or a beam would cross one of these limits. Proposals have taken this direction for neutrinos [3] and for structures around pulsars [4]. This paper tests the limits against the archives that measure them, with a null drawn from the same data for each.

1.1. Contributions

1. Tests of stellar X-ray emission against the coronal ceiling, the rotation-activity relation, the band structure of a continuum and coherent periodicity.
2. Tests of radio emission from inactive stars, of circular and linear polarisation against what stars and synchrotron radiation allow, and of spectral edges and long-period transients.

3. Tests of TeV neutrinos at nearby systems and moving with the planets, and of ultra-high-energy cosmic rays arriving in pairs.
4. Tests of photon timing, flares and pulses: ultraviolet photon clusters, flare clocks, superflare rates of planet hosts, box-shaped flares, dispersed pulses and pulsars as hosts.

2. X-rays

The coronal ceiling. A corona saturates near $L_X/L_{\rm bol}= 10^{-3}$, and flares raise it for hours; a source holding a tenth of a star's luminosity in X-rays is not a corona. The eRASS1 catalogue [5] matched to Gaia by the HamStar method [6, 7] gives 137,896 coronal counterparts with a distance and a colour, whose $L_X/L_{\rm bol}$ has percentiles 50, 90, 99 and 99.9 at -3.42, -2.85, -2.30 and -1.90 dex. Two hundred and fifty exceed 10^{-2} and none exceeds 10^{-1}. Of the 250, 153 sit off Gaia's main sequence, 38 carry a cataclysmic, young, emission-line or extragalactic type on the counterpart or on a SIMBAD object within $20''$, and 59 are main-sequence M dwarfs of 3000 to 3550 K at 50 to 630 pc, whose flares reach 10^{-2} within eRASS1's day-long average. The one K dwarf of the first pass, 1eRASS J082623.3−703142 at 7.6×10^{-2}, is the cataclysmic variable 1ES 0826−70.3, which SIMBAD files $11''$ from its Gaia counterpart.

Coronae without a dynamo. A slowly rotating star has a weak corona, so X-rays far above the rotation-activity relation of a slow rotator would need another source. For 8287 coronal counterparts of 3000 to 6000 K with rotation periods from TESS [8], Gaia DR3 and ZTF [9], the Rossby number from the turnover time of the colour's temperature [10] places each against the relation of Wright et al. [2]: the residuals scatter by 0.82 dex below a Rossby number of one, with 0.24 per cent beyond 1.5 dex. Only 26 stars reach a Rossby number of one, since the period catalogues hold few slow rotators, and the one beyond 1.5 dex is a ZTF RS CVn binary at 1 kpc.

Line-only sources. A continuum spreads its counts over the bands; an X-ray line source at 0.5 to 2 keV would hold them in one. Of the 63,043 point sources of the eRASS:3 catalogue with 80 or more counts in four bands, the largest fraction in one band has a median of 0.46 and a 99th percentile of 0.78. Holding 85 per cent in one band at 3σ with every other below 5 per cent are 13 sources at 0.2–0.5 keV, the supersoft class, 2 at 2–5 keV, the absorbed class, and none at 0.5–1 or 1–2 keV, where no continuum could put them; the same test on the released spectra is the step that remains.

Periodicities. A transmitter pulsing in X-rays at seconds to minutes would show a coherent period. In 209 Chandra detections of the 50 SIMBAD objects within 10 pc, none reaches a false-alarm probability of 10^{-3} once the dither, the frame sidebands and the red noise of flares are filtered; in 180 XMM-Newton EPIC-pn observations of stars within 20 pc, periods of 5 to 600 s reach 0.01 in four against 3.6 expected and 10^{-3} in none against 0.4.

Nearby high-energy sources. Of the 2626 Gaia sources within 20 pc, carried to eRASS1's epoch, the five above $L_X/L_{\rm bol}= 10^{-2.5}$ are M3 to M6 flare stars, and seven stars lie inside the error ellipses of unassociated Fermi-LAT sources [11] against 7.9 by chance.

3. Radio

Inactive stars. A quiet F to early K dwarf with no X-ray source and no activity should be radio-silent. Against 605 stellar radio emitters within 500 pc in VLASS [12], 30 inactive stars show 3 GHz emission, fewer than the 38 chance alignments the shifted positions give; LoTSS-DR2 [13]

holds two inactive single F to early K dwarfs and RACS-low [14] three, as many as chance gives. The two brightest coincidences are background radio galaxies: 52.6 mJy at 0.34″ from Gaia DR3 2906691964510880128 by a steady steep spectrum from 0.86 to 3 GHz that no star gives, and 81 mJy at 1.2″ from Gaia DR3 1283261877869284096, resolved by LoTSS with a second WISE source beside the star.

Circular polarisation. Coherent radio emission is circularly polarised, and in stars it comes from activity. No RACS source of 10 per cent circular polarisation at signal-to-noise six lacks a counterpart among pulsars, active stars and published brown dwarfs. Of the stars of V-LoTSS [15] and the RACS survey [16], 23 of the 31 in eRASS1's half of the sky are detected in X-rays at $L_X/L_{\rm bol}$ of 7.5×10^{-5} to 3.6×10^{-3}, the rest are RS CVn, BY Dra and flare stars by name and type, and none polarised above 30 per cent has $L_X/L_{\rm bol}$ below 10^{-5} and a period above 20 d.

The synchrotron ceiling. Synchrotron radiation in a uniform field polarises at most 72 per cent, and a source beyond it needs a coherent emitter. Of the 26,973 NVSS sources [17] above 10 mJy with polarised flux above 6 mJy, the Rician-debiased fraction's 3σ lower bound has a median of 0.058 and a 99th percentile of 0.51; the 23 above 0.72 all have polarised fluxes at or above their total fluxes, which no component can have, and lie in Fornax A's lobes and on the Galactic plane in Cygnus, Cassiopeia and Aquila, where the catalogue's polarised flux is the field's diffuse emission. None sits at a pulsar.

Spectral edges. A transmitter's band begins and ends abruptly. GLEAM-X [18] gives 27,585 sources with sixteen or more of its twenty 7.68 MHz sub-bands from 72 to 231 MHz; a step at a sub-band boundary fitted beside a curved power law gains a χ^2 whose median is 0.2 and whose 99.9th percentile is 71, the flux-scale errors common to all sources. The one source in the top thousandth with a step above 30 per cent, GLEAM-X J082817.6−221400, shares the low-frequency beam with a 1.18 Jy source 69″ away, and the sub-band fits merge and split the pair as the beam shrinks.

Long-period transients and millimetre flares. Of 16 long-period radio transients [19], ten are given a natural nature by their papers, as white-dwarf binaries, X-ray outbursts or rotators, and the other six hold no Gaia star at their dispersion-measure distances beyond the rate at control positions. Of the 66 stars of the SPT-3G millimetre flares [20], every one has an X-ray source except four M5 to M8 emission-line dwarfs at 37 to 49 pc, below eRASS1's limit.

4. Particles

Neutrinos from nearby systems. A power source or a beam of TeV neutrinos would appear as a point source in IceCube's tracks. The unbinned point-source likelihood over the ten-year sample [21] at the 271 systems within 10 pc finds none above chance, and the stack of all has $p = 0.26$; the median 90 per cent limit on an E^{-2} source is 1.6×10^{29} erg s^{-1}.

Neutrinos moving with the planets. A source on a planet moves with it. The same likelihood in each planet's frame, against 1000 time scrambles that keep the events and move the planet, gives test statistics of 0.9 for Jupiter ($p = 0.17$), 0.6 for Saturn (0.17), 0.0 for Mars (0.42) and 0.3 for Venus (0.23), where the scrambles reach 9 to 13; the 90 per cent limits on E^{-2} luminosity are 7×10^9 W at Venus, 10^{10} W at Mars, 2×10^{11} W at Jupiter and 6×10^{11} W at Saturn.

Cosmic-ray doublets. Particles launched together arrive close in direction and time. Among the 2635 events above 32 EeV of the Pierre Auger Observatory's Phase One [22], pairs within 2° number eight within ten days against a mean of 5.7 with the arrival times permuted ($p = 0.21$) and two within a day against 0.56 ($p = 0.11$), with no triplet within thirty days against 0.34 expected; one pair's circle holds Kapteyn's Star at 3.9 pc, its members 1.8 and 2.1° from it, beyond the 0.9°

resolution. In the open data, the 2355 inclined showers give seven pairs against 5.1 and the 22,751 showers of the 1500 m array 569 against 559, with 349 triplets within thirty days against 361.

5. Photons in time

Ultraviolet photon clusters. A pulsed ultraviolet source would gather photons in a few milliseconds. Of 147 GALEX near-ultraviolet eclipses of the 51 stars within 10 pc [23, 24], nine hold a cluster at a false-alarm probability below 0.01 against 4.8 from 1176 control apertures; they are five events on Wolf 359, Ross 128, Wolf 424, G 272−61 and GJ 725 A, all flare stars. GJ 725 A's is an isolated 100 ms brightening of seven photons against 0.2, clean of detector and field events and fixed to the star in rebuilt detector coordinates; a census of 116 eclipses finds 98 such bursts across the detector, at 1.4×10^{-6} per source-second, the mapped ones at the detector's edge, and none on a catalogued star like GJ 725 A's. The far-ultraviolet lists of 26 stars hold no cluster, nor do 720 control apertures, and HST's photon-event lists of 108 stars hold two below 10^{-3} against 3.7 expected.

Flare clocks. Flares come at random times, and a flare train at a fixed period would be a clock. For 2304 Kepler stars with ten or more catalogued flares [25], the H test [26] from 0.5 d to a third of the span, against twenty draws of other stars' flare times in the same quarters, finds 22 of the forty highest beating every draw away from rotation: one burst in three cycles, two at the spacecraft's quarter or downlink spacing and 19 locked in phase with scatters of 0.004 to 0.017 of the period, none below the 10^{-3} of a clock. The two long clocks on fast rotators are beats of two spot periods: KIC 7838383's 49.91 d from 1.0472 and 1.0685 d, and KIC 7746984's flare clock at half the 268 d beat of 0.7398 and 0.7377 d. Of 1426 TESS stars [27], nine of the forty highest beat every draw, eight of them bursts at the 240 d spacing of TESS's pointings, and the ninth, TIC 457100137 at 1.213 d, scatters by 0.0087 of the period.

Superflares at planet hosts. A civilization that dims or shields its star's largest flares would leave its host short of superflares. Among 60,810 TESS flares, the 83 flaring planet hosts have 28.3 per cent of their flares above 10^{34} erg among those above 10^{33}, against 35.0 ± 6.0 per cent for each host's ten nearest non-hosts in temperature, radius, distance and sectors drawn 2000 times ($p = 0.14$), and 48 per cent of them carry a flare above 10^{34} erg against 48 ± 4 per cent.

Box-shaped brightenings. A flare rises fast and decays slowly, and a switched source rises and falls alike. Among the 1806 TESS 20 s light curves of stars within 30 pc, 256 brightenings at 7σ are fitted better by a box than by a flare, against 263 dips of the same shape: the rate of instrumental steps.

Dispersed pulses. A pulse from a transmitter at a star is dispersed by the interstellar medium in front of it and by nothing behind it. In Breakthrough Listen's 349 μs products of 88 scans of the nearest stars, broadband pulses at 7σ that peak at a dispersion measure below −1, between −1 and 1, and above 1 pc cm^{-3} number 25, 36 and 9 in the scans on the stars against 39, 34 and 12 off them, so the pulses are interference; folds at the Crab, Vela, B1937+21 and J0437−4715 periods as each star sees them give 26 of 176 below $p = 0.05$, but the off-source scans of a control cadence fold as low, the red noise of the series.

Pulsars. A structure around a pulsar would re-radiate its spin-down power in the infrared. Of 1978 pulsars with positions good to 1″ [28], 143 have an AllWISE counterpart [29] against 98.6 by chance, none of them ring-like against 0.12 per shifted copy, at a sensitivity of about 2 mJy at 12 μm for a 300 K ring re-radiating 10^{32} erg s^{-1} at 1 kpc. The 45,116 stars near the antipodes of

Table 1: The tests, their samples and results.

Test	Sample	Result
Coronal ceiling	137,896 eRASS1 stars	none above 10^{-1}; the 250 above 10^{-2} explained
Coronae without a dynamo	8287 stars with periods	one beyond 1.5 dex, an RS CVn binary
Line-only X-ray sources	63,043 eRASS:3 sources	none at 0.5–2 keV
X-ray periodicities	209 Chandra, 180 XMM detections	at the chance rate
Radio from inactive stars	VLASS, LoTSS, RACS	at the chance-alignment rate
Circular polarisation	RACS, V-LoTSS	active stars, pulsars, brown dwarfs
Synchrotron ceiling	26,973 NVSS sources	field emission in all 23
Spectral edges	27,585 GLEAM-X sources	one blend
TeV neutrinos	271 systems; four planets	at background
Cosmic-ray doublets	2635 events; open data	at the chance rate
Ultraviolet photon clusters	51, 26 and 108 stars	flares
Flare clocks	2304 Kepler, 1426 TESS stars	beats, bursts and spacecraft periods
Superflares at hosts	83 hosts	as the matched stars
Dispersed pulses	88 scans	as often off the stars
Pulsars	1978 pulsars; 45,116 stars	at the chance rate

the hundred brightest pulsars, where a beacon timed to a pulsar would be aimed, hold no result at $p < 0.01$ in the coverage store.

6. Related work

Learned, Pakvasa and Zee [3] proposed neutrinos for interstellar signalling, and Osmanov [4] proposed infrared searches for structures around pulsars. The tests here rest on the surveys' own work: the eROSITA catalogues and their stellar identifications [5, 6, 7], the rotation-activity relation [1, 2], the radio surveys and their stellar samples [12, 13, 14, 15, 16, 17, 18], the review of long-period transients [19], the SPT-3G flares [20], IceCube's and the Pierre Auger Observatory's public data [21, 22] and the flare catalogues of Kepler and TESS [25, 27].

7. Conclusion

No test finds emission a star or its surroundings cannot make. The X-ray sources above the coronal ceiling are evolved, cataclysmic, young or flaring; the slow rotators sit on the rotation-activity relation; no source holds its X-rays in a band no continuum could; no nearby star pulses in X-rays. Radio emission from quiet stars arrives at the chance rate of background sources, polarised emission belongs to active stars, pulsars or the diffuse field, and no low-frequency spectrum steps. Neutrinos show no source at the nearest systems or the planets, and cosmic rays pair as chance allows. Ultraviolet photon clusters are flares, flare trains are beats and bursts, planet hosts flare as their twins do, box-shaped flares are steps, and dispersed pulses are interference.

The reach of each test is its sample's. The X-ray tests cover eRASS1's half of the sky at its depth and day-long average, so a source that flares beyond the ceiling for minutes escapes it; the neutrino

limits are 10^{22} W at the nearest systems and 10^{10} to 10^{12} W at the planets; the polarisation and spectral tests reach the surveys' flux limits; and the timing tests reach periods from milliseconds to the spans of the photon lists and light curves.

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